

Growth and properties of $\text{Cu}_2\text{ZnSnS}_4$ thin films prepared by multiple metallic layer stacks as a function of sulfurization time

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Abstract In this paper, we report the two stage growth of $\text{Cu}_2\text{ZnSnS}_4$ (CZTS) thin films as a function of sulfurization time. First, magnetron sputtered metallic precursors were deposited sequentially (Zn/Cu/Sn/Cu) over rotating glass substrates held at 230 °C. Later, the sputtered precursors were heat treated at 500 °C in the ambiance of sulfur for various time durations in the range, 10–120 min. The sulfur treated samples were examined using various analytical tools to understand the role of sulfurization time on the CZTS growth and properties. From composition and structural analysis, Zn/Cu/Sn/Cu precursors sulfurized for shorter duration (10 and 20 min) revealed severe deficiency of sulfur that resulted in several metallic, bi-metallic and metal sulfide phases. With the increase of sulfurization time to 30 min, sulfur incorporation was enhanced and reached stoichiometric ratio (~50% S) for CZTS growth, however, samples were poorly crystalline in nature and consisted of prominent Cu_{2-x}S phase as well. The Zn/Cu/Sn/Cu precursors sulfurized for 60 min exhibited prominent CZTS phase without Cu_{2-x}S phase. Further, rise in sulfurization time to 120 min enabled drastic improvement in crystallinity of CZTS phase. Raman mapping over $60\text{ }\mu\text{m} \times 60\text{ }\mu\text{m}$

for these films confirmed the homogeneous phase growth of CZTS. XPS study revealed the oxidation states of Cu^{1+} , Zn^{2+} , Sn^{4+} and S^{2-} in CZTS films. The optimized films showed high absorption coefficient of 10^5 cm^{-1} with an optical band gap of 1.51 eV. These films showed leaf like grain morphology with high mobility and low resistivity of $18.2\text{ cm}^2/\text{V-s}$ and $0.7\text{ }\Omega\text{-cm}$, respectively.

1 Introduction

The ultimate goal of any technology in photovoltaics is to achieve high efficiency combined with low processing cost and inclusion of nature abundant materials. Over the last few years, Kesterite $\text{Cu}_2\text{ZnSnS}_4$ (CZTS) materials have been researched as a potential low cost alternative to well matured CuInGaSe_2 (CIGS) and CdTe technologies because of excellent absorber suitable optoelectronic properties and moreover, all its elements are nature abundant and less toxic [1]. For the first time [2], in 1998, Ito and Nakazawa synthesized CZTS films using atom beam sputtering and reported an efficiency of 0.66%. Since then, CZTS has been thoroughly researched using variety of physical and chemical methods such as sputtering [3], evaporation [4], pulsed laser deposition [5], spray pyrolysis [6], spin coating [7], electrochemical processing [8] and several other routes [1]. The thorough research and optimization of process conditions over the decade led to significant improvement in device performance to 6.77% in 2008 using sputtered CZTS [9]. In 2013, the co-evaporated CZTS thin film solar cells have demonstrated 8.4% conversion efficiency [10] and more recently a highest conversion efficiency of 9.2% was reported [11]. Despite the persistent efforts, the CZTS's performance still remains low compared to its theoretically predicted

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efficiency of 32% [12]. In all the aforesaid reports, the device configuration of CZTS was inherited from successful CIGS technology by assuming that CZTS has similar properties as CIGS. Although, the low efficiencies of CZTS can be attributed to device architecture to some extent, but several recent reports pointing the low V_{oc} could be main limiting factor, which depends on the phase (stoichiometry) and quality of the absorber [13, 14]. Since, CZTS has quite narrow process parameter window region for achieving its thermodynamically stable phase, a slight change in growth parameters would give to rise impure secondary phases (non-stoichiometric) with poor CZTS crystalline quality [15, 16]. In two-stage preparation of CZTS, optimizing post metallic deposition conditions like sulfurization temperature, sulfur partial pressure and sulfurization time are playing vital role to avoid the formation of secondary phases [17]. There were large number of reports related to the effect of sulfurization temperature on the phase and crystal structure of the CZTS, however, a few recent reports strongly emphasizing the role of sulfurization partial pressure and time is also equally important in controlling the secondary phases [14, 17–21]. Optimization of sulfurization time is very important, which improves crystallite sizes, lowers the point defects and induces recrystallization for stable CZTS phase and thereby improvement in solar cell performances. In the above reports cited, the effect of sulfurization time is studied for CZTS films prepared by electrodeposition-sulfurization [14, 18], sol–gel sulfurization [19], solution method [21] and co-sputtering-sulfurization [20]. Moreover, there is large discrepancy in sulfurization time ranging from as low as few minutes (~5 min) to as high as 12 h [20, 22–33]. Hence, in the present study, the influence of sulfurization time on phase and crystal structure quality, composition, morphology,

opto-electrical properties of kesterite CZTS films are reported in detail. The CZTS films are prepared by sequential sputtering of metallic layers (Zn/Cu/Sn/Cu) followed by sulfur treatment at 500 °C for various time lengths. A thorough understanding of effect of stacking order and sulfurization temperature on CZTS growth was reported in our preceding works [34, 35].

2 Experimental

2.1 Materials

The analytically pure sputter targets of Cu (5N), Zn (4N) and Sn (4N) with 2'' and 3 mm thickness purchased from Testbourn Ltd., UK have been used. The sulfur powder with 99.98% was obtained from Sigma-Aldrich-414980 (Germany). The soda lime glass substrates, pure N_2 and Ar gas was obtained from VR technologies (Bengaluru). Distilled water was used for circulation over targets while operating under high DC voltage.

Table 1 Chemical composition of ZCTC layers sulfurized for different time durations

Sulfurization time (min)	Cu (at.%)	Zn (at.%)	Sn (at.%)	S (at.%)
S0	45.8	29.7	24.5	0
S10	31.6	20.8	17.1	30.5
S20	25.6	16.8	13.2	44.4
S30	22.6	15.3	12.1	50.0
S60	22.8	14.9	12.3	50.0
S120	22.9	14.9	12.2	50.0

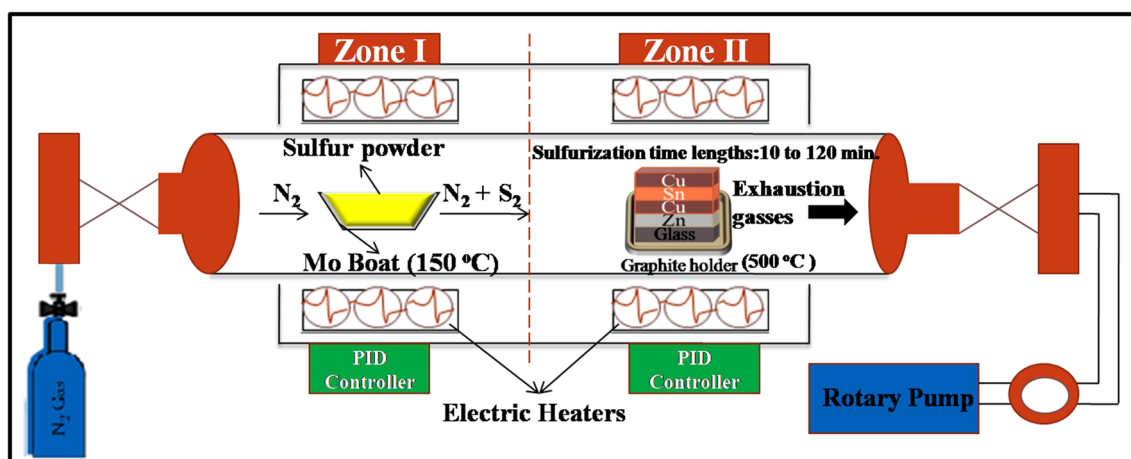


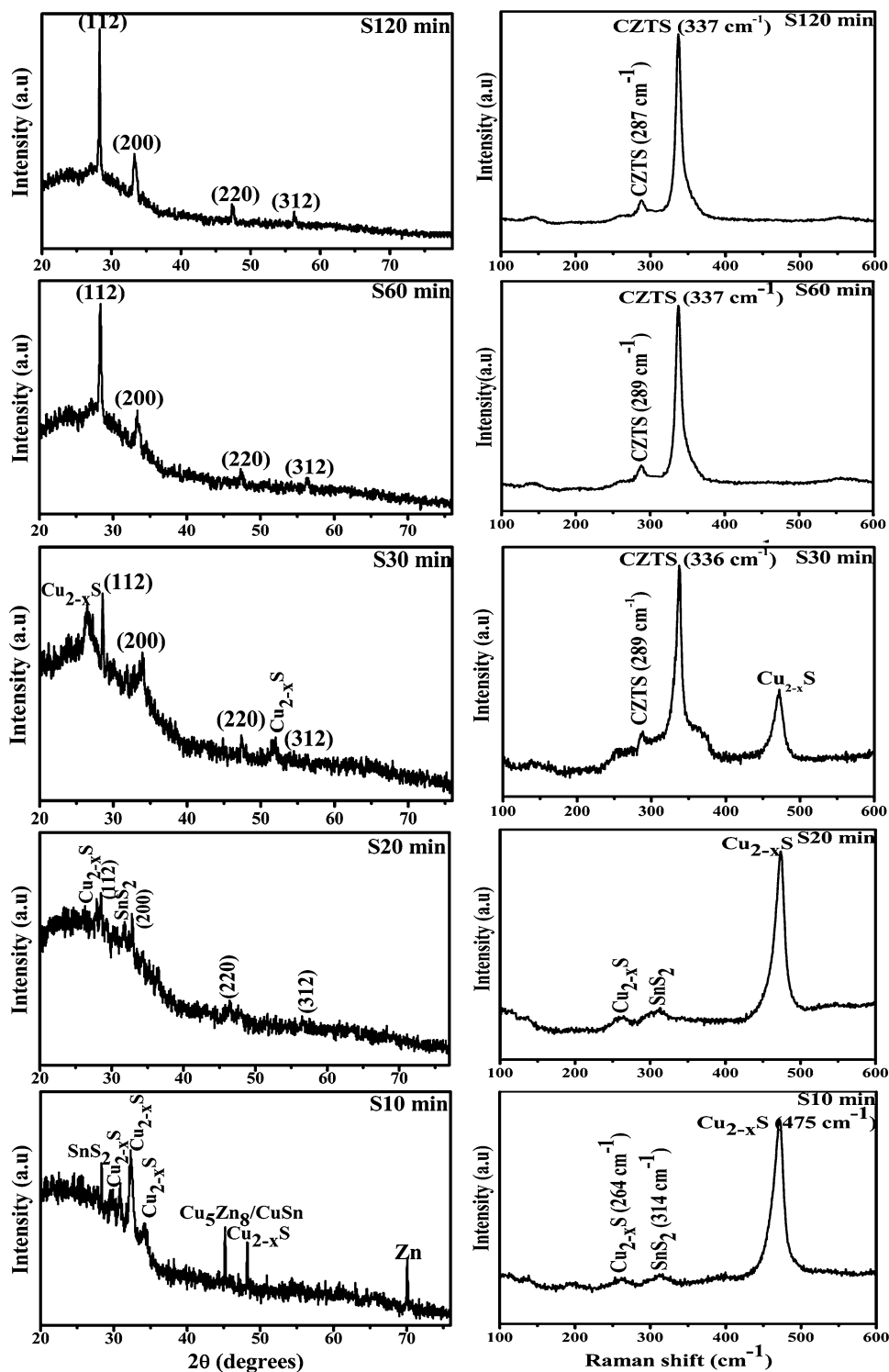
Fig. 1 Schematic depiction of sulfurization of ZCTC precursors

2.2 CZTS preparation

First, Zn/Cu/Sn/Cu (ZCTC) layers were sequentially deposited onto glass substrates held at 230 °C using DC magnetron sputtering unit (Model No: VRT SPU-06D). The detailed deposition of sequential metallic precursors and

deposition conditions have been reported in our previous studies [34, 35]. The total thickness of the ZCTC stack was approximately maintained at 300 nm by controlling the sputter time. Later, the sputtered precursors are subjected to sulfurization for different time lengths starting from 10 to 120 min using INDUFURR two zone tubular furnace

Fig. 2 X-ray diffraction (*Left*) and corresponding Raman spectra (*Right*) of ZCTC metallic stacks sulfurized for various time durations



(1.2 m long, 50 mm inner dia. and 55 mm outer dia.). Figure 1 shows schematic depiction of sulfurization of ZCTC precursors to convert into CZTS. The tubular furnace unit is consisting of two zones, on left (Zone I) and right (Zone II) of the tube, which are connected to the N_2 cylinder and rotary pump, respectively. The Mo boat loaded with sulfur powder was inserted in Zone I, while ZCTC sputtered precursors loaded in graphite container was inserted in zone II. The N_2 was purged into the quartz tube before sulfurization and approximately 20 m.bar pressure was maintained in tube throughout the sulfurization process. The sulfur source was vaporized at 150°C and N_2 carrier gas was introduced into tube with controlled flow rate. The ZCTC precursors in Zone II were heated at 500°C under sulfur ambience for different time lengths (10, 20, 30, 60 and 120 min). The precursors were then allowed for natural cooling to room temperature. After sulfurization, the samples were labeled as S10, S20, S30, S60 and S120 depending upon time length.

2.3 Characterization

The composition analysis and morphology of the sulfurized metallic precursors were examined using field emission scanning electron microscopy attached with energy dispersive X-ray analyzer (Zeiss Supra 55 VP FE-SEM). The crystalline phase and purity of CZTS was analyzed using a X-ray diffractometer (Rigaku Smart lab XRD, Cu $K\alpha$ radiation at $\lambda = 1.541 \text{ \AA}$) operated at 40 kV/200 mA and a Raman spectrometer (LabRAM HR and WITec, Alpha 300 S) operated with 488/532 nm laser. Raman Mapping was collected using scanning near field optical microscope (SNOM). Agilent Cary 5000 UV-Vis-NIR was used to record transmittance versus wavelength

spectra in the range, 200–2500 nm. X-ray photo electron spectroscopy (Kratos XPS Ultra DLD) was used to measure the valance states of Cu, Zn, Sn and S in CZTS films. Electrical studies were carried out using hot probe and Hall effect setup. An Optical profilometer (Zeta-3D) was used to measure the thickness of the sulfurized films.

3 Results and discussion

3.1 Composition analysis

Table 1 shows elemental composition of ZCTC stack films deposited on glass and sulfurized for various time lengths. Before sulfurization, the precursor stacks have shown slightly Cu-poor and Zn-rich composition, which is highly suitable composition for the growth of pure CZTS as per literature. The Cu-poor composition is known to enhance the shallow acceptors formation like V_{Cu} defects whereas the Zn-rich composition is required to suppress deep acceptors like Cu_{Zn} defects. Further, at this composition relatively less number of detrimental $[\text{Cu}_{\text{Zn}} + \text{Sn}_{\text{Zn}}]$ and $[2\text{Cu}_{\text{Zn}} + \text{Sn}_{\text{Zn}}]$ defect clusters are formed, which favors pure CZTS growth [36, 37]. At shorter time lengths (10 and 20 min), the samples are found to be severe sulfur deficient (~ 30 and 44% S), which resulted metallic, bi-metallic and metal sulfide phases. With the increase of sulfurization time to 30 min, sulfur incorporation into samples was enhanced and reached close to stoichiometric value of $\sim 50\%$ S required for CZTS growth. The composition remains unchanged with further increase in sulfurization time to 120 min.

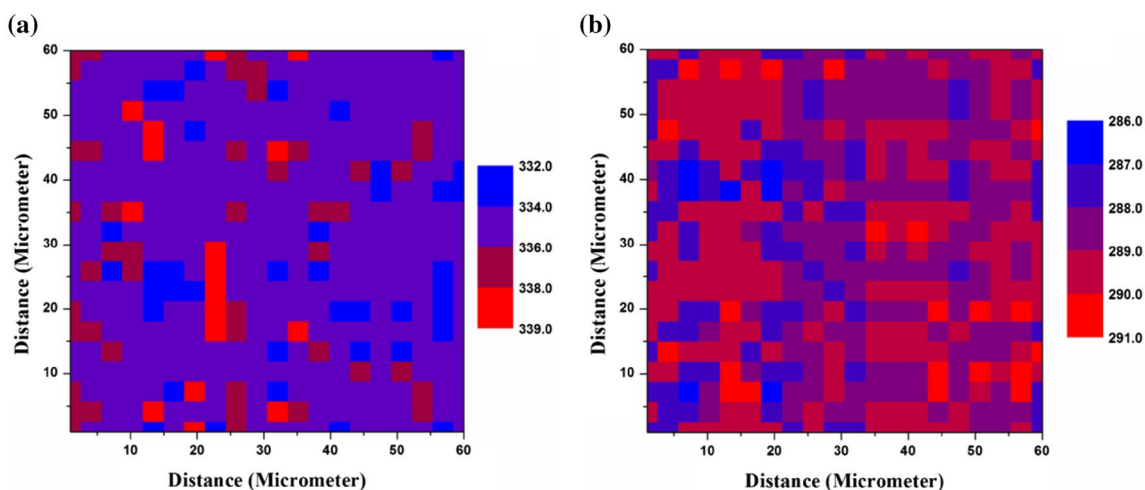


Fig. 3 Raman mapping for **a** strong mode and **b** weak mode of CZTS in samples treated for 120 min

3.2 Structural studies

Figure 2 shows X-ray diffraction and Raman spectra of ZCTC metallic stacks sulfurized for various time durations. Prior to CZTS formation, the intermediate ZnS and Cu_2SnS_3 phases are often formed, which show similar diffraction pattern in XRD. Hence, Raman spectroscopic studies were often conducted to distinguish them as a separate phase and also confirm the growth of CZTS phase [38]. A broad hump in XRD spectra around $20\text{--}25^\circ$ can

be attributed to glass substrates. It is observed from XRD and Raman analysis that the stacks sulfurized for 10 min had mainly metallic (Zn), bi-metallic ($\text{Cu}_5\text{Zn}_8/\text{CuSn}$), and metal sulfides (Cu_{2-x}S and SnS_2) phases. The formation of metallic and metal sulfide phases is due to severe sulfur deficiency resulting from insufficient and short sulfurization time for homogeneous mixing of metallic precursors with sulfur i.e. incomplete sulfurization. It is well known that at initial stages of CZTS growth, Cu_{2-x}S phase is likely to be formed dominantly due to its lower formation energy

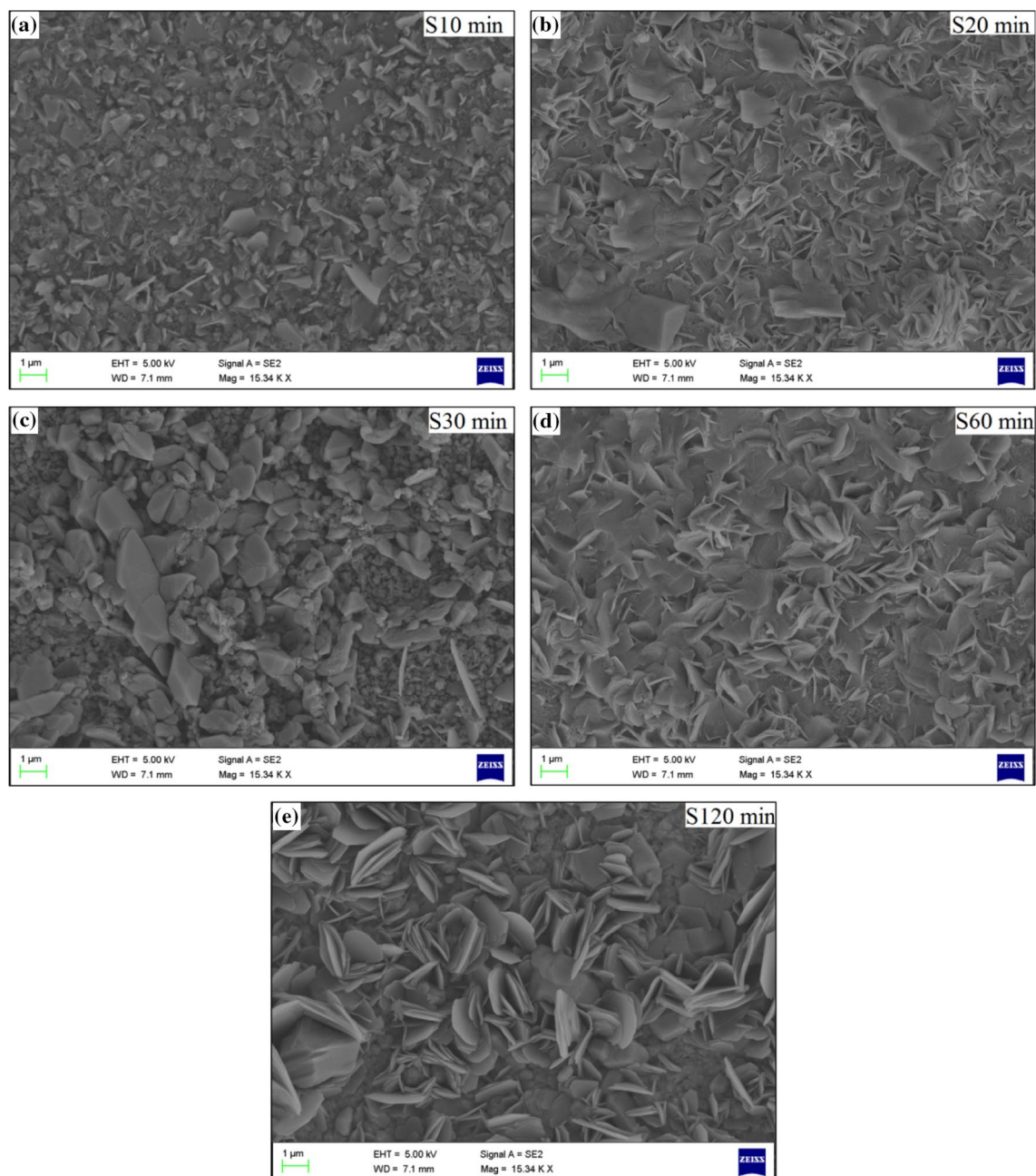


Fig. 4 a–e FE-SEM images of ZCTC precursor stack sulfurized at different time durations

when compared to all other phases of Cu–Zn–Sn–S system [39]. At 20 min of sulfurization time, the peaks related to SnS_2 , Cu_{2-x}S along with onset of CZTS diffraction peaks at 28.5 , 32.9 , 46.4 and 58.2° was observed [40]. However, in corresponding Raman spectra any modes of CZTS were not found due to minor CZTS phase compared to Cu_{2-x}S phase. At 30 min of sulfurization time, the observed diffraction pattern can be perfectly attributed to prominent $\text{Cu}_2\text{ZnSnS}_4$ phase, as its corresponding Raman spectra exhibits strong modes of CZTS (289 and 336 cm^{-1}) along with a minor Cu_{2-x}S phase (475 cm^{-1}). The films sulfurized for 60 min yield prominent CZTS phase and it seems that all other secondary impurities are consumed to form a single phase CZTS. Further, the longer sulfurization time (120 min) leads to enhancement in the intensity of the X-ray diffraction (112) peak and particularly CZTS Raman strong mode at 336 cm^{-1} and decrease of its full width half maximum (FWHM) from 9.5 to 5.5 cm^{-1} indicating highly crystalline CZTS growth. The crystallite size calculated from (112) diffraction plane was increased from 17 to 56 nm and its corresponding FWHM was decreased from 0.5 to 0.15 , further strengthening the quality of the grown CZTS.

Interestingly, we observed that sulfur residence time in Zone I was only for about half an hour, within that time all sulfur was vaporized and transported to, in and around the film Zone II. So we believe that, in total sulfurization time, except first half an hour, the rest of the time is simply vacuum annealing in sulfur ambience in zone II, which is also important to initiate reactions among impure phases (binaries and ternaries) if remain, and consume to form a final single phase CZTS. The prolonged vacuum annealing is essential to avoid secondary phase formation and further helps in lowering defects and improving crystallinity of CZTS [18]. From XRD and Raman analysis, it is concluded that longer sulfurization time of 120 min is required for homogeneous mixing, inter diffusion and to grow defect free and compact grain CZTS. Further, Raman mapping was collected for strong

and weak modes over $60\text{ }\mu\text{m} \times 60\text{ }\mu\text{m}$ on precursors sulfurized for 120 min and is shown in Fig. 3. The $60\text{ }\mu\text{m} \times 60\text{ }\mu\text{m}$ mapping shows wavenumber values corresponding to strong ($332\text{--}339\text{ cm}^{-1}$) and weak modes ($286\text{--}291\text{ cm}^{-1}$) of CZTS, found good agreement with literature [40], indicating highly uniform growth of CZTS.

3.3 Morphology

The morphology of the ZCTC metallic stacks sulfurized for different time durations are shown in Fig. 4. At low sulfurization time periods upto 60 min, the small sized non uniform grains were seen. However, with increase of sulfurization time ($>60\text{ min}$) the grains grew in size and finally at 120 min of sulfurization time, leaf like structures with clear and bigger compact grain morphology were observed. The improved morphology is understood to be due to homogeneous mixing of all elements, which induces high crystallinity as evidenced in XRD and Raman spectra for longer sulfurization time duration. The larger grains could lessen the grain boundaries that would enhance minority carrier diffusion lengths and built in potential, which are prime prerequisites for high efficiency thin film solar cells [41].

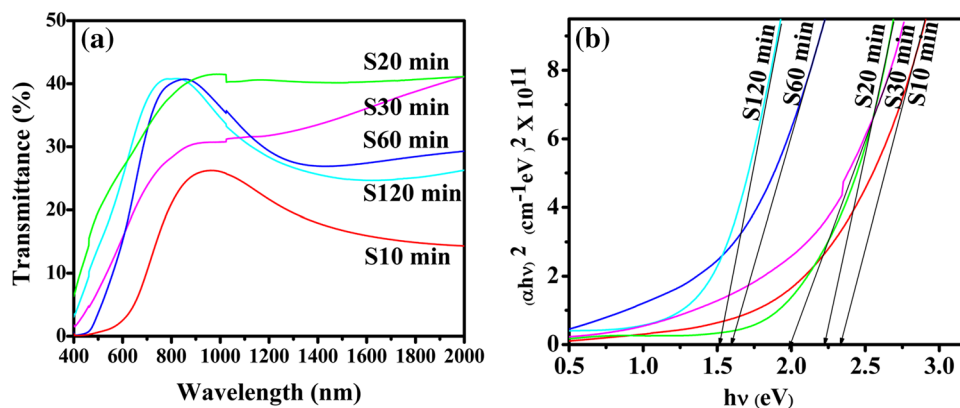
3.4 Optical properties

Figure 5a shows transmittance spectra as a function of wavelength for precursor films sulfurized for various time lengths. The sudden drop of transmittance below 820 nm indicating the onset of material absorption. Absorption coefficient ' α ' from transmittance spectra was calculated using the following Equations [42],

$$\alpha = \frac{-\ln T}{t}$$

where ' T ' is transmittance and ' t ' is the thickness of the film, $0.8\text{ }\mu\text{m}$, which is measured using profilometer.

Fig. 5 **a** Transmittance spectra as a function of wavelength **b** $(\alpha h\nu)^2$ versus $h\nu$ plot for ZCTC stack sulfurized for various time durations



Absorption coefficient was found to be in the range of $2 \times 10^5 \text{ cm}^{-1}$ for all the sulfurized films. Energy band gap was calculated tauc relations [43],

$$(\alpha h\nu)^n = A(h\nu - E_g)$$

where ‘n’ values are 2, 2/3, 1/2 and 1/4 for direct allowed, direct forbidden, indirect allowed and indirect forbidden transitions respectively. In the present case, $n=2$ was

satisfied for the films sulfurized at various timings indicating that direct allowed transition of all phases present in the films. Figure 5b shows the estimation of optical band gap values for different sulfurization time durations. The estimated optical band gaps for metallic stack sulfurized at 10, 20 and 30 min were 2.34, 2.20 and 1.90 eV respectively. The expected secondary phases like tetragonal Cu_2SnS_3 have a low band gap of 1.15 eV [44], while SnS_2 has higher band gap of 2.12 eV [45] when compared

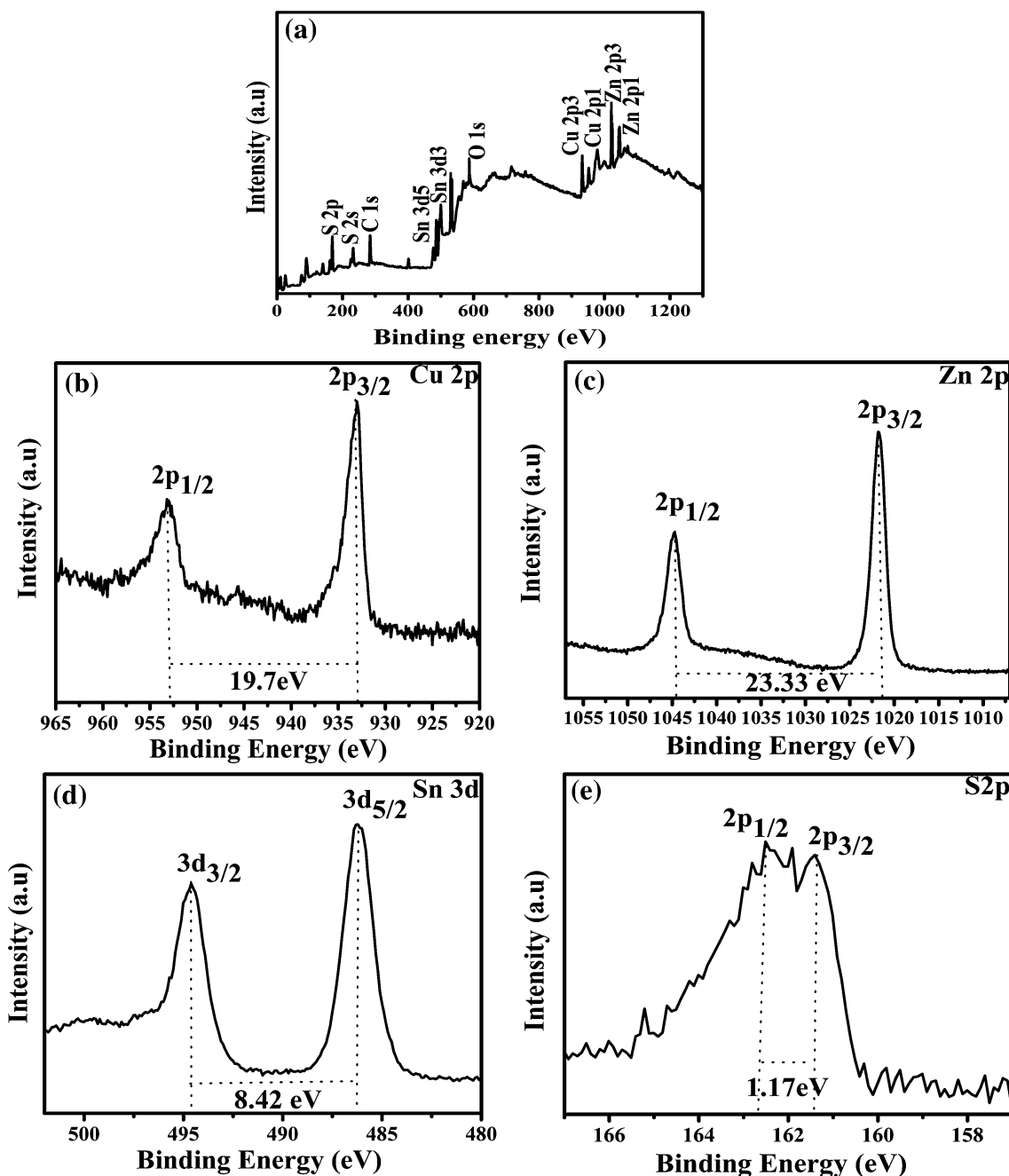


Fig. 6 a Survey scan of XPS spectra and high resolution core level spectra for b Cu 2p, c Zn 2p, d Sn 3d and e S 2p of the ZCTC metallic stack sulfurized at 120 min

to CZTS (1.5 eV). The band gap of Cu_{2-x}S varies from 1.1 to 2.16 eV where 'x' is a variable ranging from 0 to 1 [46, 47]. The observed variation in the band gap for low sulfurization timings can be attributed to the combination of secondary phases. The films sulfurized for 60 min have shown band gap of 1.58 eV while as the films sulfurized at 120 min had 1.51 eV. This small decrease of band gap with sulfurization time might be due to increased crystallinity. The valance band is made up by anti-bonding linear combination of Cu-3d states and S-3p states and the conduction band is dominated by an isolated band made up by Sn-5s and S-3p states. The excitation of carriers across the gap of 1.51 eV from the Cu-3d/S-3p states into Sn-5s/S-3p conduction band determines the electrical properties of CZTS [48, 49].

3.5 XPS analysis

The valance states of Cu, Zn, Sn and S in ZCTC metallic stack sulfurized at 120 min were determined by X-ray photo electron spectroscopy. Figure 6 shows over-all survey scan of precursors sulfurized for 120 min and its corresponding core level spectrum for various constituents. In Cu 2p spectra, the doublet peaks at 932.00 eV (Cu 2p_{3/2}) and 951.80 eV (Cu 2p_{1/2}) with a separation 19.8 eV confirm the +1 state of Cu in CZTS [50]. The peaks appeared at 1022.34 eV (Zn 2p_{3/2}) and 1045.34 eV (Zn 2p_{1/2}) with a separation 23.0 eV confirms the +2 state of Zn in CZTS films. Chemical environment of Zn in CZTS is different than that in ZnS. The absence of peaks at 1021.2 and 1044.2 eV ruled out the existence of ZnS and hence indicated that the film has pure CZTS phase [51, 52]. The peak separation of 8.5 eV between two well resolved peaks at binding energies of 486.13 eV (3d_{5/2}) and 494.55 eV (3d_{3/2}) reveals +4 state of Sn in CZTS films [53]. In Sulfur 2p spectra, two peaks located at 161.41 eV (2p_{3/2}) and 162.58 eV (2p_{1/2}) with a splitting of 1.2 eV demonstrate the existence of S in -2 oxidation state [54]. Therefore, the observed oxidation states of Cu¹⁺, Zn²⁺, Sn⁴⁺ and S²⁻ confirms the formation of pure phase CZTS.

3.6 Electrical studies

Hot probe test demonstrated that all the films were p-type in nature. The room temperature electrical properties of the films sulfurized for 30, 60 and 120 min are given in Table 2. Resistivity and hole mobility mainly depends on the morphology of the films and the presence of secondary phases. The stacks sulfurized for 30 min showed high carrier concentration and low mobility, due to admixture of semi-metallic p-type nature of Cu_{2-x}S phase [55]. The films sulfurized for 60 and 120 min exhibited nearly same carrier concentrations in the order of 10^{17} cm^{-3} , however,

Table 2 Room temperature hall measurements of ZCTC stacks sulfurized for 30, 60 and 120 min time lengths

Sample	Resistivity ($\Omega\text{-cm}$)	Carrier concentration (cm^{-3})	Mobility ($\text{cm}^2/\text{V-s}$)
S30	2.7	1.2×10^{18}	1.9
S60	1.5	4.1×10^{17}	10.2
S120	0.7	4.9×10^{17}	18.2

the stacks sulfurized at 120 min showed lower resistivity of $0.7 \Omega\text{-cm}$ and high mobility of $18.2 \text{ cm}^2/\text{V-s}$, which can be accounted to improved crystallinity when compared to the other films sulfurized for different timings.

4 Conclusions

We have systematically investigated the effect of sulfurization time on the growth and properties of CZTS thin film absorber. The structural, chemical composition, morphological and electrical properties of the ZCTC precursor films after the sulfurization for various time durations were presented. The ZCTC precursors sulfurized for shorter time periods (10 and 20 min) showed metallic, bi-metallic and metal sulfide phases due to sulfur deficiency. Sulfurization to 30 min duration, shows the onset of CZTS with prominent Cu_{2-x}S phase. Metallic precursors sulfurized for 60 min exhibited prominent CZTS phase without any secondary phases. Further, rise in sulfurization time to 120 min exhibited highly crystalline CZTS with dominant (112) orientation. Raman mapping over $60 \mu\text{m} \times 60 \mu\text{m}$ shows the films were highly uniform. XPS study confirms the oxidation states of Cu¹⁺, Zn²⁺, Sn⁴⁺ and S²⁻ in pure phase CZTS films sulfurized for 120 min. High sulfurization timings of about 120 min is required for homogeneous mixing, inter diffusion and to grow defect free and compact grain CZTS from metallic precursor's stack. The observed opto-electronic properties indicate its suitability and potential use as absorber material in thin film solar cells.

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